

# 20260302EB\_MN (NCT, RNA)

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**2026年3月2日 星期一**

split iPSCs into a full 6-well plate on the Friday before differentiation

differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates

keep some iPSCs

**2026年3月4日 星期三**

T2 -> T3

**2026年3月6日 星期五**

T3 -> T4 (~14 mL for over-weekend)

**2026年3月9日 星期一**

T4 -> T5

**2026年3月11日 星期三**

T5 -> T6

**2026年3月13日 星期五**

T6 -> T6 (~14mL for over-weekend)

**2026年3月16日 星期一**

T6 -> T7

Dissociation (process dishA and dishB separately)

1. Collect EBs into 50mL tubes (each plate). Wash the plate with 5mL PBS and added them into the tubes. Let EBs settle with gravity.
2. Remove sup. and add 4mL 0.25% trypsin (prepare 16mL 0.25% trypsin with 400U DNaseI)
3. Pipette ~8-10 with P1000 and then incubate the tubes in 37C.
4. pipette\*10 at 5min, 10min, 15min, and 20min
5. add 8mL FBS into 50mL tube -> pipette with 5mL pipet ~ 10 times
6. add 8mL CTWM -> pipette ~ 10 times
7. filter cells through 40um cell strainer into new tubes.
8. centrifuge at 200xg, 4min

| cell info1 |      |         |       |                        |
|------------|------|---------|-------|------------------------|
|            | A    | B       | C     | D                      |
| 1          | cell | passage | well# |                        |
| 2          | WT11 | P2      | 4     | for both batch A and B |

| Table2 |             |             |
|--------|-------------|-------------|
|        | A           | B           |
| 1      |             | volume (mL) |
| 2      | trypsin     | 4           |
| 3      | HI-FBS/CTWM | 8           |

| Table3 |        |              |          |              |                 |        |            |                     |                 |              |
|--------|--------|--------------|----------|--------------|-----------------|--------|------------|---------------------|-----------------|--------------|
|        | A      | B            | C        | D            | E               | F      | G          | H                   | I               | J            |
| 1      |        | conc. (M/mL) | survival | total volume | tota cell count | plate# | well plate | cell count/well (K) | number of wells | batch        |
| 2      | dish A | 1.44         | 100      | 5            | 7.2             | I      | 24         | 225                 | 12              | from batch A |
| 3      | dish B | 1.2          | 98       | 5            | 6               | I      | 24         | 225                 | 12              | from batch B |
| 4      |        |              |          |              |                 | II     | 6 (A1-3)   | 1125                | 3               | batch A      |
| 5      |        |              |          |              |                 |        | 6 (B1-3)   | 1650                | 3               | batch B      |

| Plate I |                              |                              |                      |                              |                              |                      |
|---------|------------------------------|------------------------------|----------------------|------------------------------|------------------------------|----------------------|
|         | 1                            | 2                            | 3                    | 4                            | 5                            | 6                    |
| A       | 110<br>(12uL<br>in<br>total) | 111<br>(18uL<br>in<br>total) | 6uL of<br>vLH1<br>85 | 110<br>(12uL<br>in<br>total) | 111<br>(18uL<br>in<br>total) | 6uL of<br>vLH1<br>85 |
| B       |                              |                              |                      |                              |                              |                      |
| C       |                              |                              | 8uL of<br>vLH1<br>85 |                              |                              | 8uL of<br>vLH1<br>85 |
| D       |                              |                              |                      |                              |                              |                      |

| Plate II |         |   |   | ^ |
|----------|---------|---|---|---|
|          | 1       | 2 | 3 |   |
| A        | batch A |   |   |   |
| B        | batch B |   |   |   |

2026年3月17日 星期二

Change T7 -> T7 if added virus

**\*\*Note:** cells on glass bottom plate din't look well. A lot of cell death and cells sparsed and streched out. It might because the virus was too much? It was okay on plate II  
However, P2 iPSC-derived MNs have attached poorly three times -> try not to use P2 iPSCs in the future.

2026年3月18日 星期三

T7-> T7 if not added virus

2026年3月19日 星期四

T7 -> T8

2026年3月20日 星期五

1/2 T8 -> 1/2 T8

2026年3月23日 星期一

1/2 T8 -> 1/2 T8

- ☒ Treat 24h for batch A NCT
- ☐ Prepare more no-gluc media
- ☐ Treat RLG Day1

RLG

Media needed for RLG

2.5mM gluc:  $9 \times 4(\text{day}) \times 2\text{mL} = 72\text{mL}$   
0.15mM gluc:  $3 \times 4(\text{day}) \times 2\text{mL} = 24\text{mL}$   
5.5mM gluc:  $6 \times 4(\text{day}) \times 2\text{mL} = 48\text{mL}$

2026年3月25日 星期三

1/2 T8 -> 1/2 T8

Image NCT (1h and 24h)

1. Aliquote 850 uL and 820 uL into 1.5mL eppendorf.

2. Add SirDNA (1:5000) with treatment (0.5 uL into 10 uL media. Add 2uL into 500uL media on the plate for 24h treatment. Add 6 uL into 1500uL media for 1h treatment)

Well1 ^

|   | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| A | 1 | 1 |   | 1 | 1 | 1 |
| B | 2 | 2 |   | 2 | 2 | 2 |
| C | 3 | 3 | 1 | 3 | 3 |   |
| D | 4 | 4 | 2 | 4 | 4 |   |

NCT\_O.C. , 60X, 600x900, 16-bit ^

|   | A       | B   | C         | D             | E      | F                                         |
|---|---------|-----|-----------|---------------|--------|-------------------------------------------|
| 1 | channel | bit | power (%) | exposure time | note   |                                           |
| 2 |         | 561 | 16        | 30            | 300 ms | the signal for 110 is good. 111 was okay. |
| 3 |         | 640 | 16        | 15            | 100 ms |                                           |
| 4 |         | 488 | 16        | 50            | 1s     | for 185, use 35% power. Signal was bad.   |

time table for NCT treatment ^

|    | A            | B   | C                  | D           | E          | F       | G             | H                                                                 | I                                                                                                            | J            | K  | L                  | M           | N          | O       | P             | Q  | R                                                                                |
|----|--------------|-----|--------------------|-------------|------------|---------|---------------|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|--------------|----|--------------------|-------------|------------|---------|---------------|----|----------------------------------------------------------------------------------|
| 1  | batch A      |     |                    |             |            |         |               |                                                                   |                                                                                                              | batch B      |    |                    |             |            |         |               |    |                                                                                  |
| 2  | condition#   |     | condition          | media       | time       | imaging | imaging order |                                                                   |                                                                                                              | condition#   |    | condition          | media       | time       | imaging | imaging order |    |                                                                                  |
| 3  | on 3/25/2026 |     | 110                |             |            |         |               | note                                                              | note2                                                                                                        | on 3/27/2026 |    | 110                |             |            |         |               |    |                                                                                  |
| 4  | 1            | A 1 | 2.5mM glucose 1h   | 2.5mM gluc  | 9:25       | 10:23   | 1             |                                                                   |                                                                                                              | 1            | A4 | 2.5mM glucose 1h   | 2.5mM gluc  | 12:05      | 13:05   |               | 1  |                                                                                  |
| 5  | 2            | B 1 | 0.15mM glucose 1h  | 0.15mM gluc | 9:45       | 10:51   | 2             |                                                                   |                                                                                                              | 2            | B4 | 0.15mM glucose 1h  | 0.15mM gluc | 12:40      | 12:40   |               | 2  |                                                                                  |
| 6  | 3            | C 1 | 2.5mM glucose 24h  | 2.5mM gluc  | 3/24 14:55 | 14:49   | 7             |                                                                   |                                                                                                              | 3            | C4 | 2.5mM glucose 24h  | 2.5mM gluc  | 3/26 16:00 | 16:30   |               | 5  |                                                                                  |
| 7  | 4            | D 1 | 0.15mM glucose 24h | 0.15mM gluc | 3/24 14:55 | 15:22   | 8             | the 561 laser (30%) was also on when taking 640...                |                                                                                                              | 4            | D4 | 0.15mM glucose 24h | 0.15mM gluc | 3/26 16:00 | 17:15   |               | 6  |                                                                                  |
| 8  |              |     | 111                |             |            |         |               |                                                                   |                                                                                                              |              |    | 111                |             |            |         |               |    |                                                                                  |
| 9  | 1            | A 2 | 2.5mM glucose 1h   | 2.5mM gluc  | 10:00      | 11:15   | 3             |                                                                   |                                                                                                              | 1            | A5 | 2.5mM glucose 1h   | 2.5mM gluc  | 13:20      | 14:20   |               | 3  |                                                                                  |
| 10 | 2            | B 2 | 0.15mM glucose 1h  | 0.15mM gluc | 10:15      | 11:45   | 4             | the time interval of treatment was wrong... Significantly delayed |                                                                                                              | 2            | B5 | 0.15mM glucose 1h  | 0.15mM gluc | 14:03      | 15:03   |               | 4  |                                                                                  |
| 11 | 3            | C 2 | 2.5mM glucose 24h  | 2.5mM gluc  | 3/24 16:00 | 15:54   | 9             |                                                                   |                                                                                                              | 3            | C5 | 2.5mM glucose 24h  | 2.5mM gluc  | 3/26 17:00 | 17:45   |               | 7  |                                                                                  |
| 12 | 4            | D 2 | 0.15mM glucose 24h | 0.15mM gluc | 3/24 16:00 | 16:23   | 10            | 561 laser was off for 640 channel                                 |                                                                                                              | 4            | D5 | 0.15mM glucose 24h | 0.15mM gluc | 3/26 17:00 | 18:15   |               | 8  |                                                                                  |
| 13 |              |     | 185                |             |            |         |               |                                                                   |                                                                                                              |              |    | 185                |             |            |         |               |    |                                                                                  |
| 14 | 1            | C 3 | 2.5mM glucose 1h   | 2.5mM gluc  | 12:40      | 13:40   | 5             | a lot of debris. But A3 and B3 have low signal.                   | Most signal is evenly distributed in nucleus and cytosol. It's hard to find cells with lower nuclear signal. | 1            | A6 | 2.5mM glucose 1h   | 2.5mM gluc  | 18:01      | 19:01   |               | 9  | I tried to use cells with less virus since the previous batch looked not healthy |
| 15 | 2            | D 3 | 0.15mM glucose 1h  | 0.15mM gluc | 13:10      | 14:15   | 6             |                                                                   |                                                                                                              | 2            | B6 | 0.15mM glucose 1h  | 0.15mM gluc | 18:30      | 19:30   |               | 10 |                                                                                  |

| 20260325_fileID_treatment_NCT |                                       |          |             |      |           |           |      |
|-------------------------------|---------------------------------------|----------|-------------|------|-----------|-----------|------|
|                               | A                                     | B        | C           | D    | E         | F         | G    |
| 1                             | image                                 | reporter | treatment   | time | type      | replicate | note |
| 2                             | com_110_MN_no-glu_50%_1s_30s_60x_0001 | 110      | 2.5mM gluc  | 1h   | no-change | A         |      |
| 3                             | com_110_MN_no-glu_50%_1s_30s_60x_0002 | 110      | 2.5mM gluc  | 1h   | no-change | A         |      |
| 4                             | com_110_MN_no-glu_50%_1s_30s_60x_0003 | 110      | 0.15mM gluc | 1h   | no-change | A         |      |
| 5                             | com_110_MN_no-glu_50%_1s_30s_60x_0004 | 110      | 0.15mM gluc | 1h   | no-change | A         |      |
| 6                             | com_110_MN_no-glu_50%_1s_30s_60x_0005 | 110      | 2.5mM gluc  | 24h  | no-change | A         |      |
| 7                             | com_110_MN_no-glu_50%_1s_30s_60x_0006 | 110      | 2.5mM gluc  | 24h  | no-change | A         |      |
| 8                             | com_110_MN_no-glu_50%_1s_30s_60x_0007 | 110      | 0.15mM gluc | 24h  | no-change | A         |      |
| 9                             | com_110_MN_no-glu_50%_1s_30s_60x_0008 | 110      | 0.15mM gluc | 24h  | no-change | A         |      |
| 10                            | com_111_MN_no-glu_50%_1s_30s_60x_0001 | 111      | 2.5mM gluc  | 1h   | no-change | A         |      |
| 11                            | com_111_MN_no-glu_50%_1s_30s_60x_0002 | 111      | 2.5mM gluc  | 1h   | no-change | A         |      |
| 12                            | com_111_MN_no-glu_50%_1s_30s_60x_0003 | 111      | 0.15mM gluc | 1h   | no-change | A         |      |
| 13                            | com_111_MN_no-glu_50%_1s_30s_60x_0004 | 111      | 0.15mM gluc | 1h   | no-change | A         |      |
| 14                            | com_111_MN_no-glu_50%_1s_30s_60x_0005 | 111      | 2.5mM gluc  | 24h  | no-change | A         |      |
| 15                            | com_111_MN_no-glu_50%_1s_30s_60x_0006 | 111      | 2.5mM gluc  | 24h  | no-change | A         |      |
| 16                            | com_111_MN_no-glu_50%_1s_30s_60x_0007 | 111      | 0.15mM gluc | 24h  | no-change | A         |      |
| 17                            | com_111_MN_no-glu_50%_1s_30s_60x_0008 | 111      | 0.15mM gluc | 24h  | no-change | A         |      |
| 18                            | com_185_MN_no-glu_50%_1s_30s_60x_0001 | 185      | 2.5mM gluc  | 1h   | no-change | A         |      |
| 19                            | com_185_MN_no-glu_50%_1s_30s_60x_0002 | 185      | 2.5mM gluc  | 1h   | no-change | A         |      |
| 20                            | com_185_MN_no-glu_50%_1s_30s_60x_0003 | 185      | 0.15mM gluc | 1h   | no-change | A         |      |
| 21                            | com_185_MN_no-glu_50%_1s_30s_60x_0004 | 185      | 0.15mM gluc | 1h   | no-change | A         |      |

| out of focus frames |                                       |       |       |       |       |    |    |    |    |
|---------------------|---------------------------------------|-------|-------|-------|-------|----|----|----|----|
|                     | A                                     | B     | C     | D     | E     | F  | G  | H  | I  |
| 1                   |                                       | 1     | 2     | 3     | 4     | 5  | 6  | 7  | 8  |
| 2                   | com_111_MN_no-glu_50%_1s_30s_60x_0002 | 16-18 | 18-22 | 20-23 | 22-25 | 23 | 23 | 23 | 23 |

\*\*\*\*\*I meessed up the time interval for the NCT experiments.... It should be 30min in between but I did 15min....

2026年3月27日 星期五

- I prepared fresh 2.5mM gluc NBA full and used the remaining 0.15mM gluc NBA full

| 20260327_fileID_treatment_NCT |                                       |          |             |      |           |           |      |
|-------------------------------|---------------------------------------|----------|-------------|------|-----------|-----------|------|
|                               | A                                     | B        | C           | D    | E         | F         | G    |
| 1                             | image                                 | reporter | treatment   | time | type      | replicate | note |
| 2                             | com_110_MN_no-glu_50%_1s_30s_60x_0001 | 110      | 2.5mM gluc  | 1h   | no-change | B         |      |
| 3                             | com_110_MN_no-glu_50%_1s_30s_60x_0002 | 110      | 2.5mM gluc  | 1h   | no-change | B         |      |
| 4                             | com_110_MN_no-glu_50%_1s_30s_60x_0003 | 110      | 0.15mM gluc | 1h   | no-change | B         |      |
| 5                             | com_110_MN_no-glu_50%_1s_30s_60x_0004 | 110      | 0.15mM gluc | 1h   | no-change | B         |      |
| 6                             | com_110_MN_no-glu_50%_1s_30s_60x_0005 | 110      | 2.5mM gluc  | 24h  | no-change | B         |      |
| 7                             | com_110_MN_no-glu_50%_1s_30s_60x_0006 | 110      | 2.5mM gluc  | 24h  | no-change | B         |      |
| 8                             | com_110_MN_no-glu_50%_1s_30s_60x_0007 | 110      | 0.15mM gluc | 24h  | no-change | B         |      |
| 9                             | com_110_MN_no-glu_50%_1s_30s_60x_0008 | 110      | 0.15mM gluc | 24h  | no-change | B         |      |
| 10                            | com_111_MN_no-glu_50%_1s_30s_60x_0001 | 111      | 2.5mM gluc  | 1h   | no-change | B         |      |
| 11                            | com_111_MN_no-glu_50%_1s_30s_60x_0002 | 111      | 2.5mM gluc  | 1h   | no-change | B         |      |
| 12                            | com_111_MN_no-glu_50%_1s_30s_60x_0003 | 111      | 0.15mM gluc | 1h   | no-change | B         |      |
| 13                            | com_111_MN_no-glu_50%_1s_30s_60x_0004 | 111      | 0.15mM gluc | 1h   | no-change | B         |      |
| 14                            | com_111_MN_no-glu_50%_1s_30s_60x_0005 | 111      | 2.5mM gluc  | 24h  | no-change | B         |      |
| 15                            | com_111_MN_no-glu_50%_1s_30s_60x_0006 | 111      | 2.5mM gluc  | 24h  | no-change | B         |      |
| 16                            | com_111_MN_no-glu_50%_1s_30s_60x_0007 | 111      | 0.15mM gluc | 24h  | no-change | B         |      |
| 17                            | com_111_MN_no-glu_50%_1s_30s_60x_0008 | 111      | 0.15mM gluc | 24h  | no-change | B         |      |
| 18                            | com_185_MN_no-glu_50%_1s_30s_60x_0001 | 185      | 2.5mM gluc  | 1h   | no-change | B         |      |
| 19                            | com_185_MN_no-glu_50%_1s_30s_60x_0002 | 185      | 2.5mM gluc  | 1h   | no-change | B         |      |
| 20                            | com_185_MN_no-glu_50%_1s_30s_60x_0003 | 185      | 0.15mM gluc | 1h   | no-change | B         |      |
| 21                            | com_185_MN_no-glu_50%_1s_30s_60x_0004 | 185      | 0.15mM gluc | 1h   | no-change | B         |      |

RNA

- Treat cells for RNA extraction 18:40-19:40
- 1. remove media
  - 2. Add 1mL Trizol
  - 3. collect samples into 1.5mL tube and freeze them in -20 (according to the Trizol protocol. Samples can be stored in -20 for up to a year. Since our -80 freezer warmed up and could not be used, I used -20 instead)

| treatment for Plate II |               |                |                |
|------------------------|---------------|----------------|----------------|
|                        | 1             | 2              | 3              |
| A                      | 2.5mM gluc 1h | 2.5mM gluc 1h  | 0.15mM gluc 1h |
| B                      | 2.5mM gluc 1h | 0.15mM gluc 1h | 0.15mM gluc 1h |